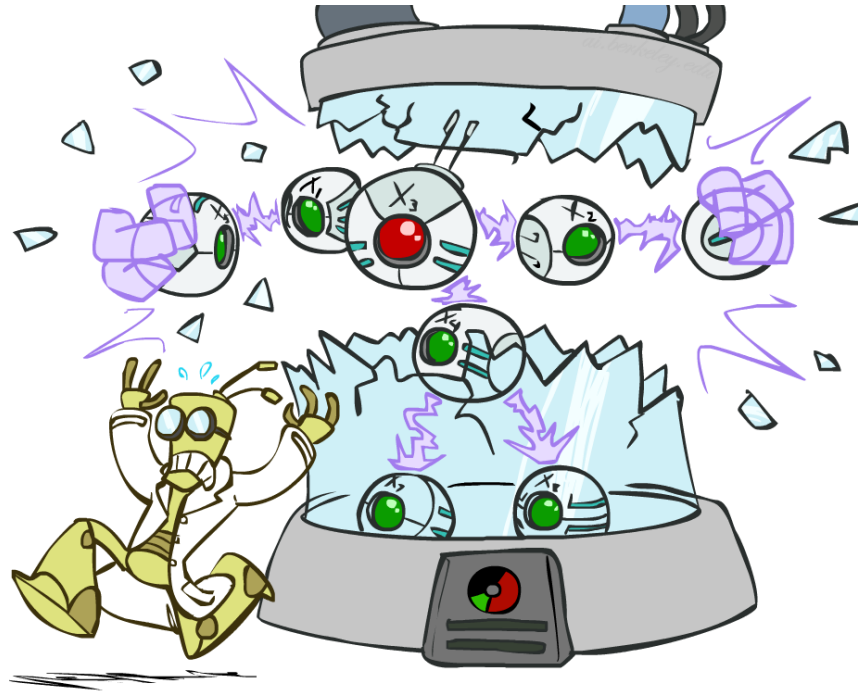


# CS 6300: Artificial Intelligence

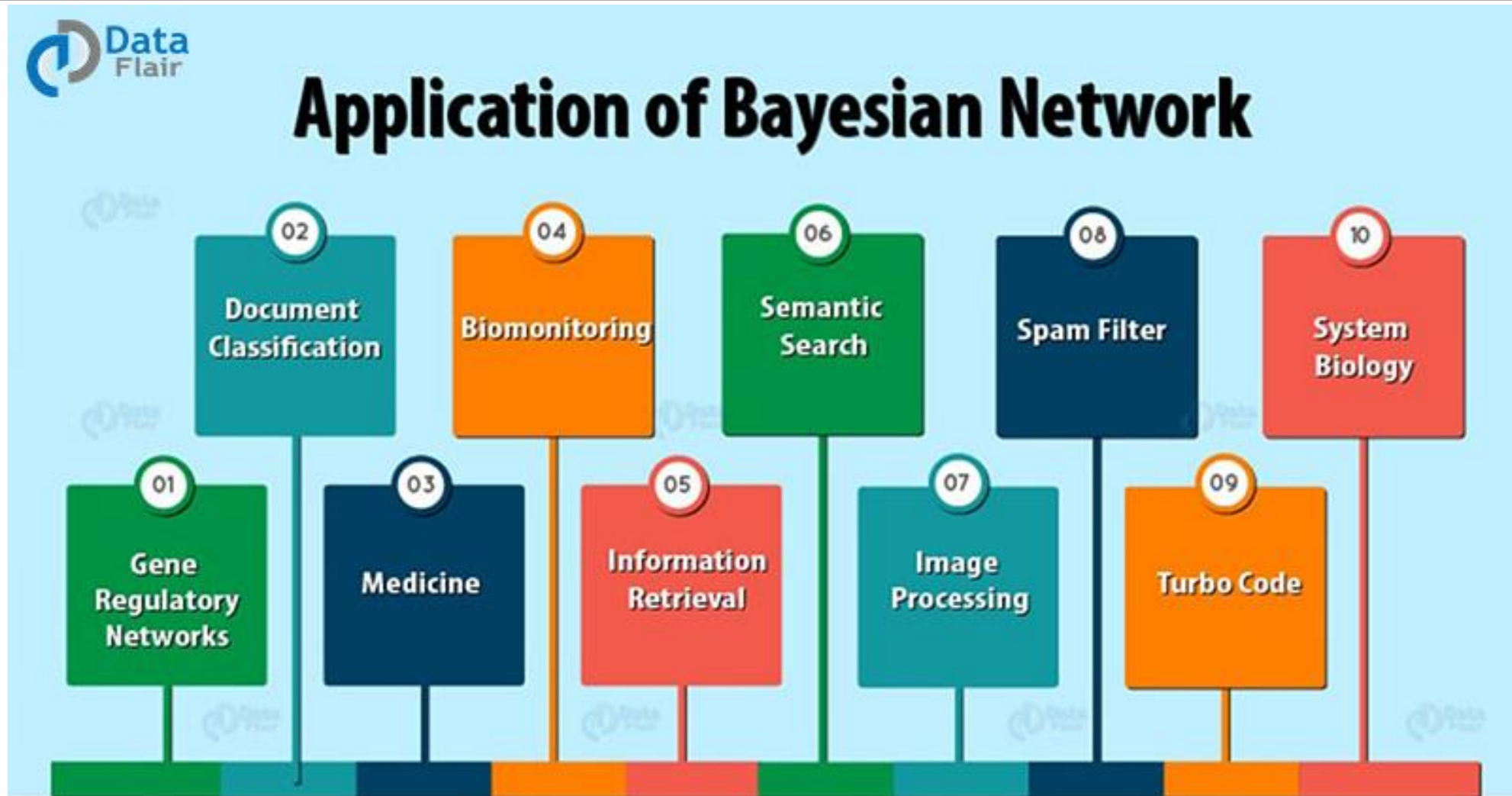
## Bayes' Nets: Independence and Basic Inference



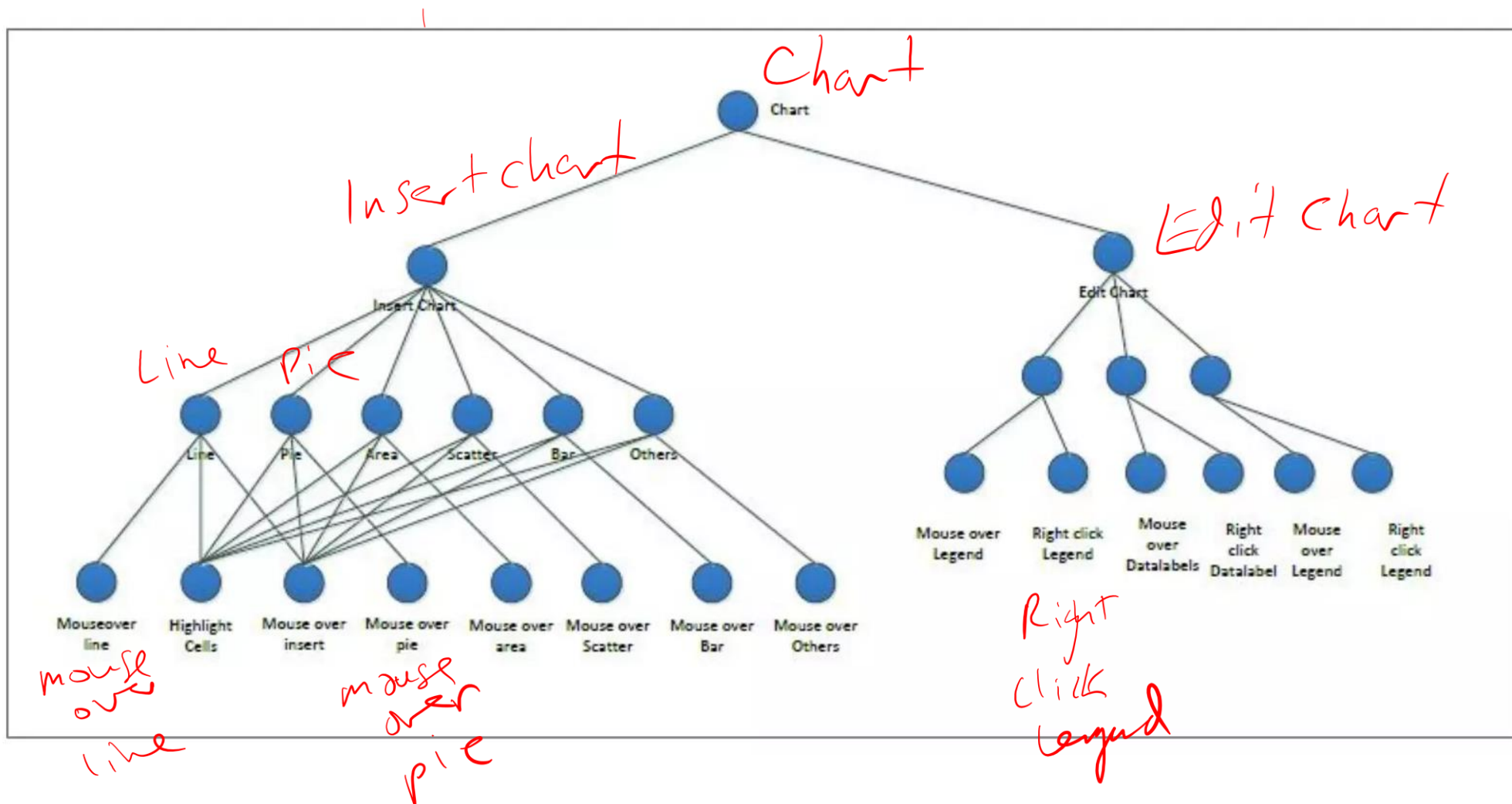
Instructor: Daniel Brown --- University of Utah

[Based on slides created by Dan Klein and Pieter Abbeel <http://ai.berkeley.edu>.]

# Where are Bayes' Nets used?



# (Old-school) Bayes' Net Application: Clippy



It looks like you're writing a letter.

Would you like help?

- Get help with writing the letter
- Just type the letter without help

☐ Don't show me this tip again



# Bayes' Net Application: Spam Filtering

- Input: an email
- Output: spam/ham
- Setup:
  - Get a large collection of example emails, each labeled "spam" or "ham"
  - Note: someone has to hand label all this data!
  - Want to learn to predict labels of new, future emails
- Features: The attributes used to make the ham / spam decision
  - Words: FREE!
  - Text Patterns: \$dd, CAPS
  - Non-text: SenderInContacts
  - ...



Dear Sir.

First, I must solicit your confidence in this transaction, this is by virtue of its nature as being utterly confidential and top secret. ...



TO BE REMOVED FROM FUTURE MAILINGS, SIMPLY REPLY TO THIS MESSAGE AND PUT "REMOVE" IN THE SUBJECT.

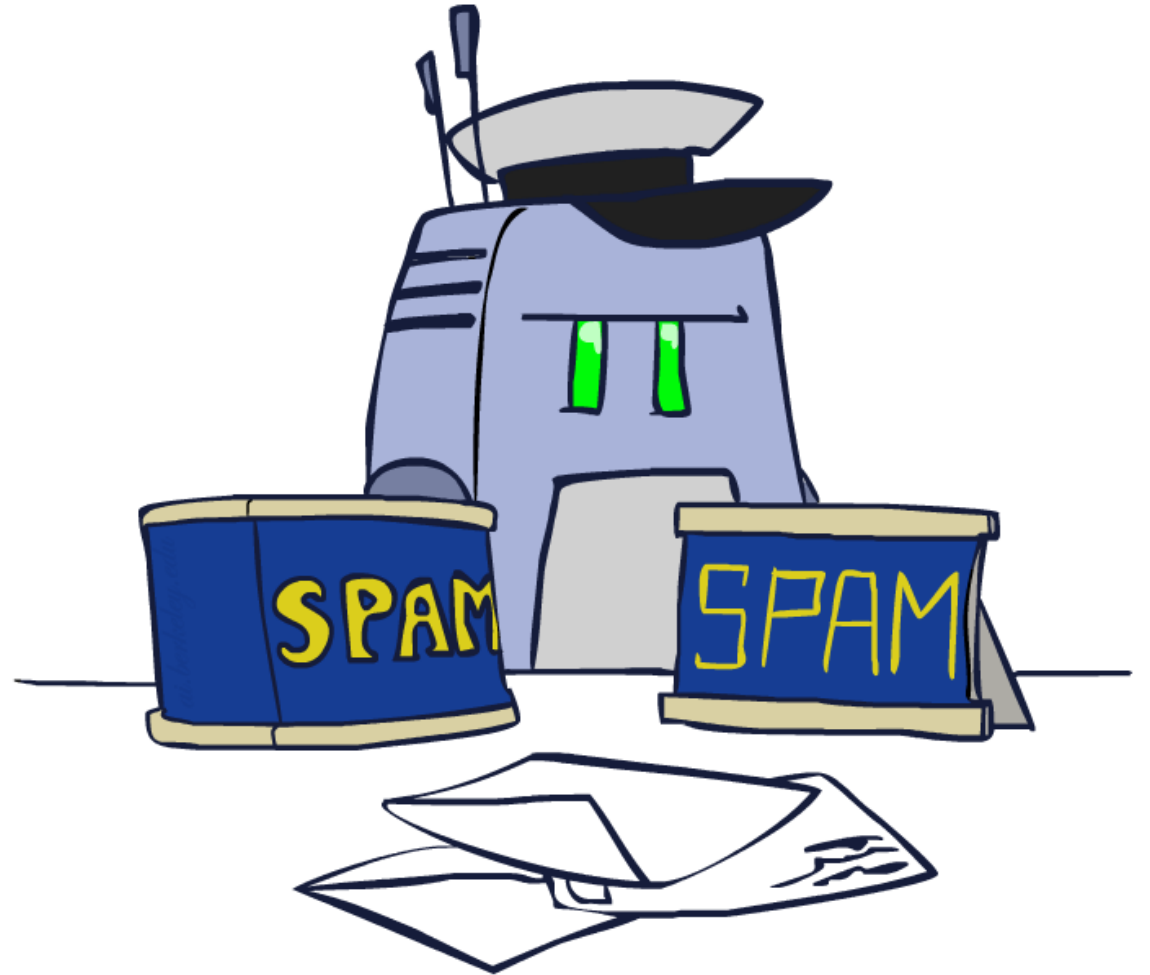
99 MILLION EMAIL ADDRESSES  
FOR ONLY \$99



Ok, I know this is blatantly OT but I'm beginning to go insane. Had an old Dell Dimension XPS sitting in the corner and decided to put it to use, I know it was working pre being stuck in the corner, but when I plugged it in, hit the power nothing happened.

# Bayes' Net Application: Spam Filtering

- Model-based approach
  - Build a model (e.g. Bayes' net) where both the label and features are random variables
  - Instantiate any observed features
  - Query for the distribution of the label conditioned on the features
- Challenges
  - What structure should the BN have?
  - How should we learn its parameters?



# Naïve Bayes

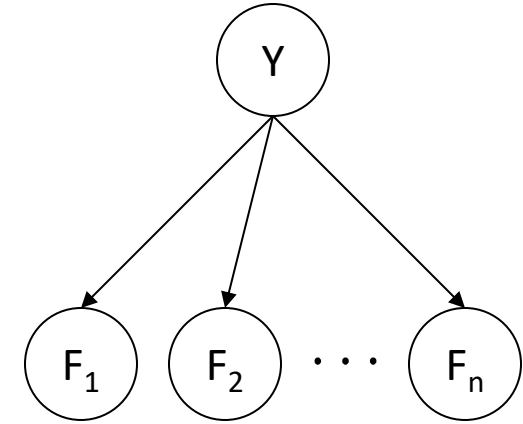
- A general Naive Bayes model:

$|Y|$  parameters

$$P(Y, F_1 \dots F_n) = P(Y) \prod_i P(F_i|Y)$$

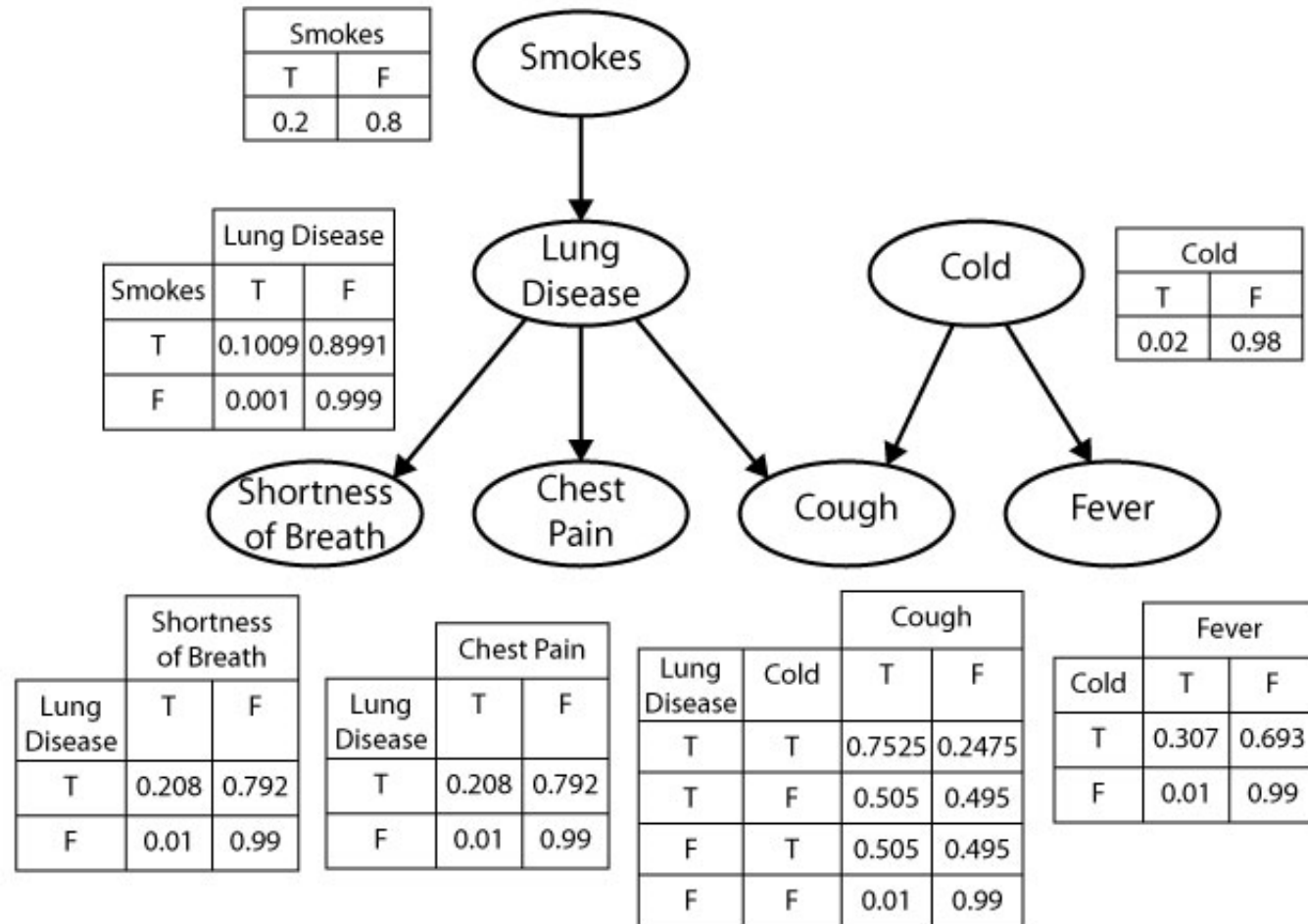
$|Y| \times |F|^n$  values

$n \times |F| \times |Y|$   
parameters

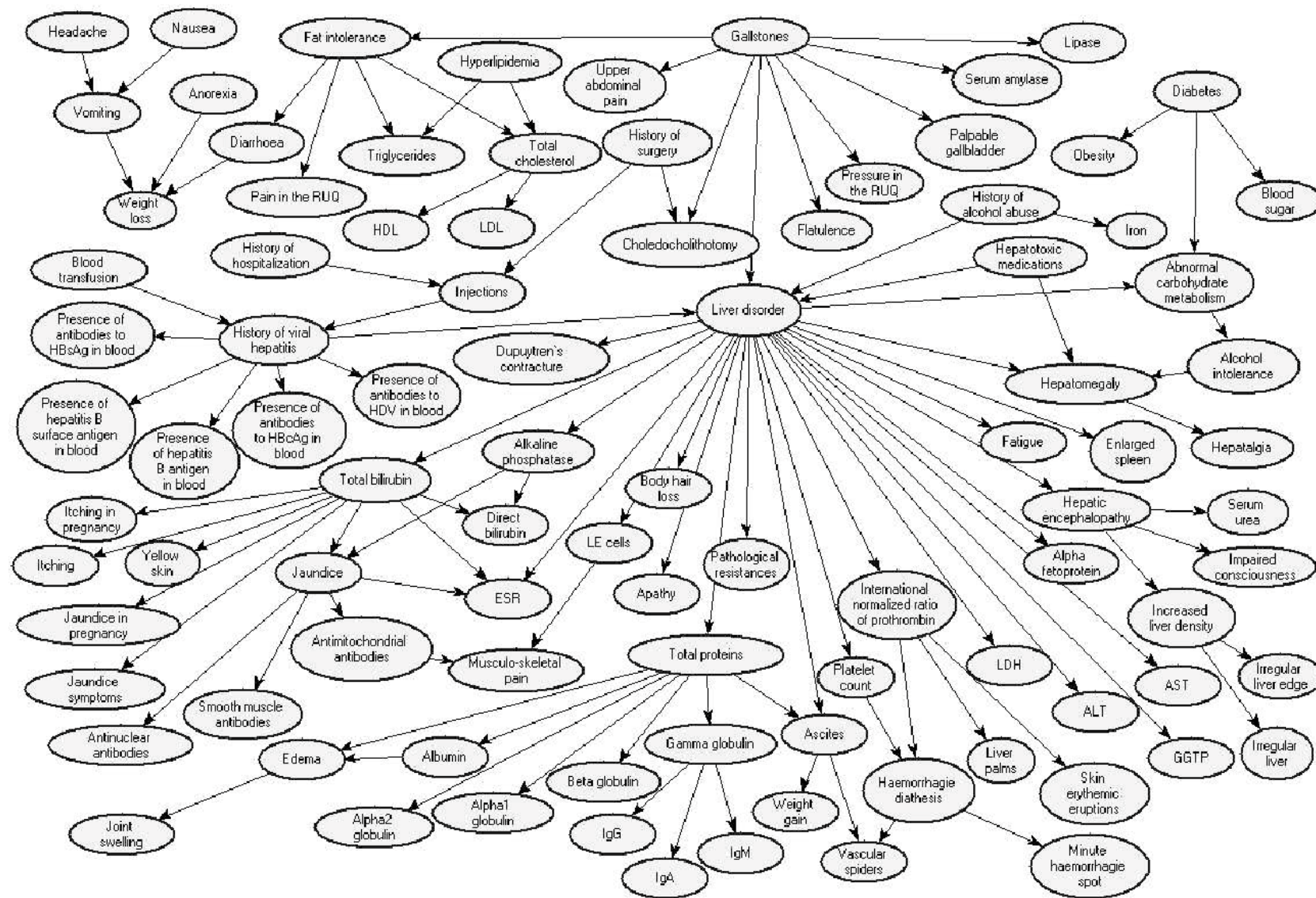


- We only have to specify how each feature depends on the class
- Total number of parameters is *linear* in  $n$
- Model is very simplistic, but often works anyway
- Assumes all features are independent effects of the label  $Y$  (very Naïve, but very efficient)

# Medical Diagnosis









# Probability Recap

- Conditional probability

$$P(x|y) = \frac{P(x, y)}{P(y)}$$

## Bayes' Rule

$$P(x|y) = \frac{P(y|x)P(x)}{P(y)}$$

- Product rule

$$P(x, y) = P(x|y)P(y)$$

- Chain rule

$$\begin{aligned} P(X_1, X_2, \dots, X_n) &= P(X_1)P(X_2|X_1)P(X_3|X_1, X_2) \dots \\ &= \prod_{i=1}^n P(X_i|X_1, \dots, X_{i-1}) \end{aligned}$$

- X, Y independent if and only if:  $\forall x, y : P(x, y) = P(x)P(y)$



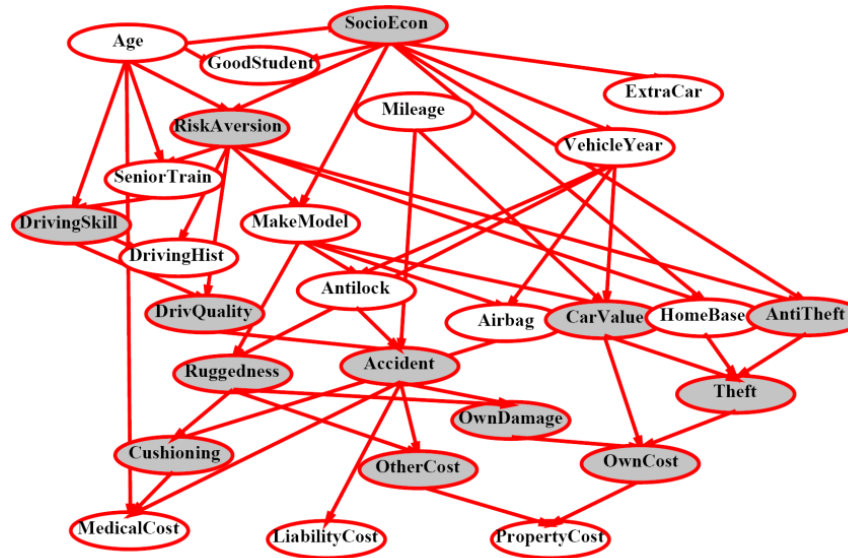
- X and Y are conditionally independent given Z if and only if:

$$X \perp\!\!\!\perp Y | Z$$

$$\forall x, y, z : P(x, y|z) = P(x|z)P(y|z)$$

# Bayes' Nets

- A Bayes' net is an efficient encoding of a probabilistic model of a domain



- Questions we can ask:
  - Inference: given a fixed BN, what is  $P(X \mid e)$ ?
  - Representation: given a BN graph, what kinds of distributions can it encode?
  - Modeling: what BN is most appropriate for a given domain?

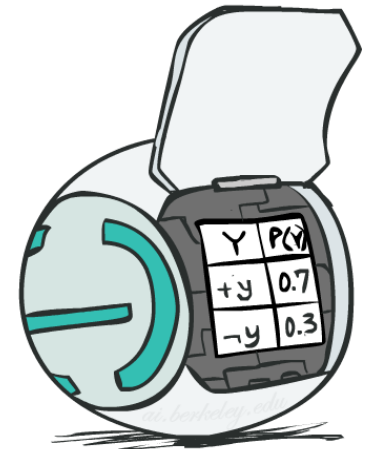
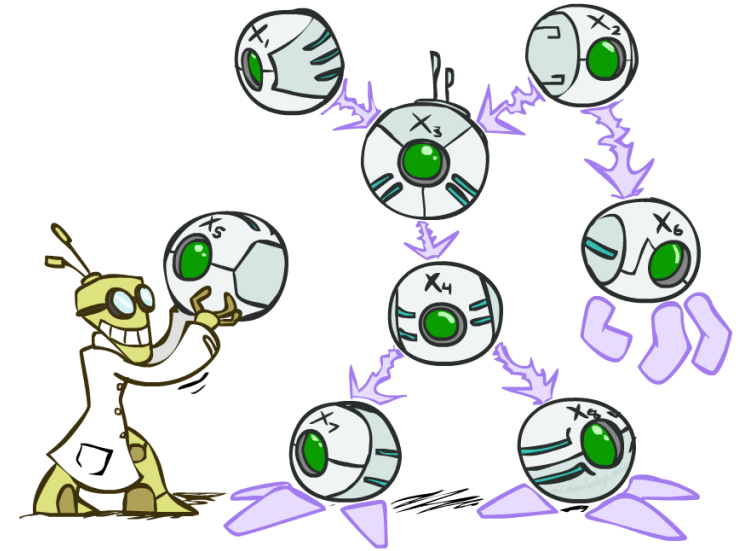
# Bayes' Net Semantics

- A directed, acyclic graph, one node per random variable
- A conditional probability table (CPT) for each node
  - A collection of distributions over  $X$ , one for each combination of parents' values

$$P(X|a_1 \dots a_n)$$

- Bayes' nets implicitly encode joint distributions
  - As a product of local conditional distributions
  - To see what probability a BN gives to a full assignment, multiply all the relevant conditionals together:

$$P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i))$$



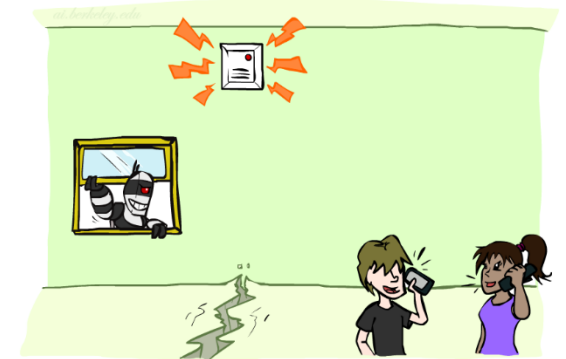
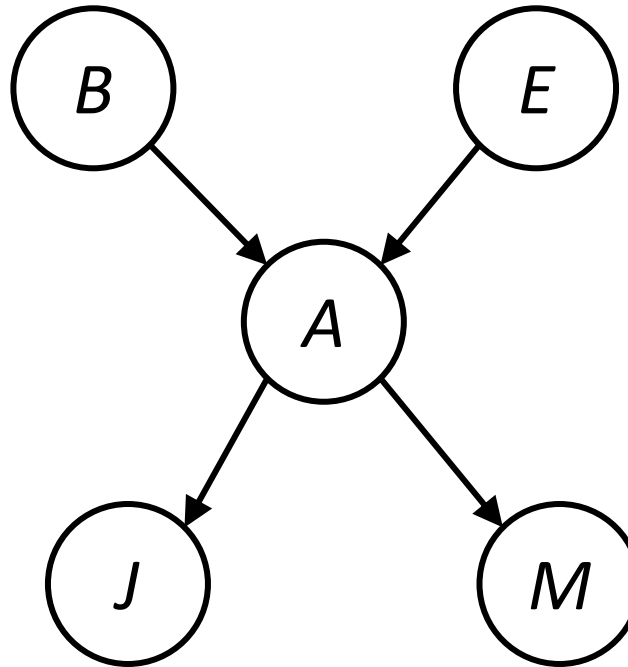
# Example: Alarm Network

| B  | P(B)  |
|----|-------|
| +b | 0.001 |
| -b | 0.999 |

| E  | P(E)  |
|----|-------|
| +e | 0.002 |
| -e | 0.998 |

| A  | J  | P(J A) |
|----|----|--------|
| +a | +j | 0.9    |
| +a | -j | 0.1    |
| -a | +j | 0.05   |
| -a | -j | 0.95   |

| A  | M  | P(M A) |
|----|----|--------|
| +a | +m | 0.7    |
| +a | -m | 0.3    |
| -a | +m | 0.01   |
| -a | -m | 0.99   |



$$P(+b, -e, +a, -j, +m) =$$

| B  | E  | A  | P(A B,E) |
|----|----|----|----------|
| +b | +e | +a | 0.95     |
| +b | +e | -a | 0.05     |
| +b | -e | +a | 0.94     |
| +b | -e | -a | 0.06     |
| -b | +e | +a | 0.29     |
| -b | +e | -a | 0.71     |
| -b | -e | +a | 0.001    |
| -b | -e | -a | 0.999    |

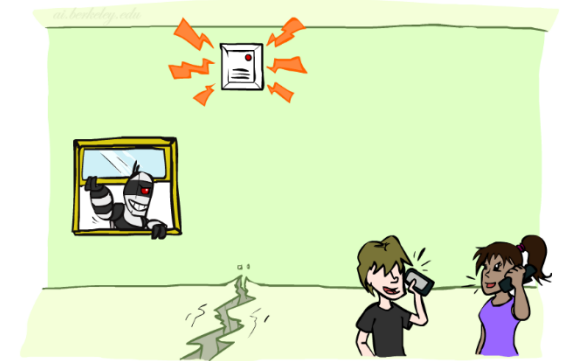
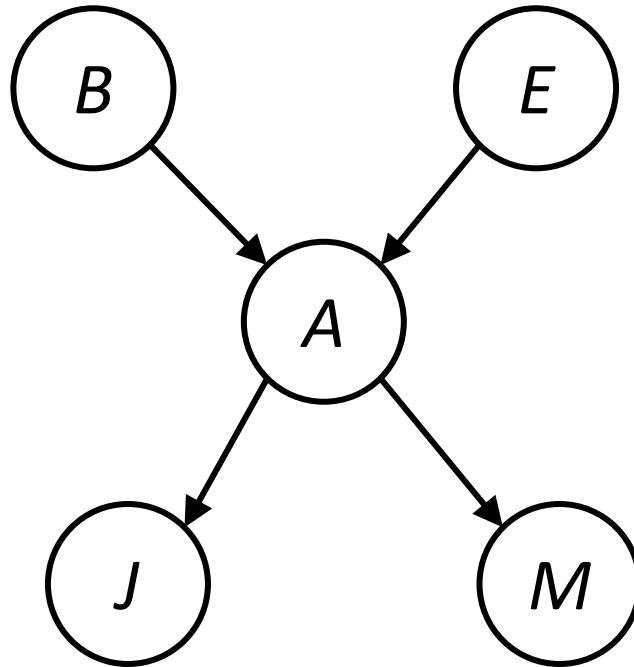
# Example: Alarm Network

| B  | P(B)  |
|----|-------|
| +b | 0.001 |
| -b | 0.999 |

| E  | P(E)  |
|----|-------|
| +e | 0.002 |
| -e | 0.998 |

| A  | J  | P(J A) |
|----|----|--------|
| +a | +j | 0.9    |
| +a | -j | 0.1    |
| -a | +j | 0.05   |
| -a | -j | 0.95   |

| A  | M  | P(M A) |
|----|----|--------|
| +a | +m | 0.7    |
| +a | -m | 0.3    |
| -a | +m | 0.01   |
| -a | -m | 0.99   |



| B  | E  | A  | P(A B,E) |
|----|----|----|----------|
| +b | +e | +a | 0.95     |
| +b | +e | -a | 0.05     |
| +b | -e | +a | 0.94     |
| +b | -e | -a | 0.06     |
| -b | +e | +a | 0.29     |
| -b | +e | -a | 0.71     |
| -b | -e | +a | 0.001    |
| -b | -e | -a | 0.999    |

$$\begin{aligned}
 P(+b, -e, +a, -j, +m) &= \\
 P(+b)P(-e)P(+a|+b, -e)P(-j|+a)P(+m|+a) &= \\
 0.001 \times 0.998 \times 0.94 \times 0.1 \times 0.7 &
 \end{aligned}$$

# Size of a Bayes' Net

- How big is a joint distribution over  $N$  Boolean variables?

$$2^N$$

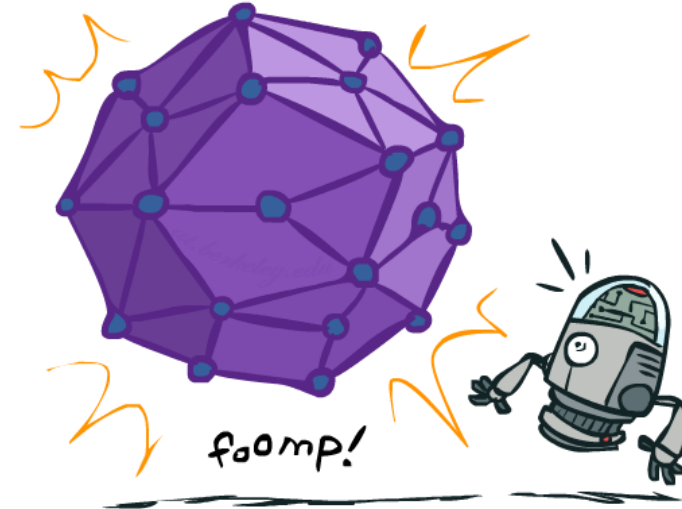
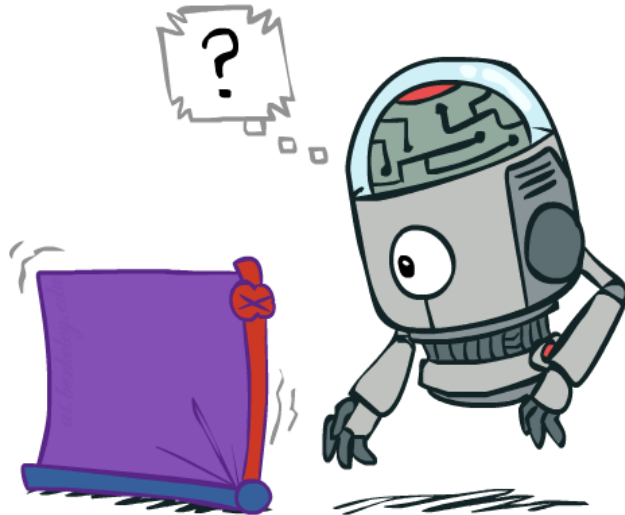
- How big is an  $N$ -node net if nodes have up to  $k$  parents?

$$O(N * 2^{k+1})$$

- Both give you the power to calculate

$$P(X_1, X_2, \dots, X_n)$$

- BNs: Huge space savings!
- Also easier to elicit local CPTs
- Also faster to answer queries (coming)





# Bayes' Nets

---

## Representation

- Conditional Independences
- Probabilistic Inference
- Learning Bayes' Nets from Data

# Conditional Independence

- X and Y are **independent** if

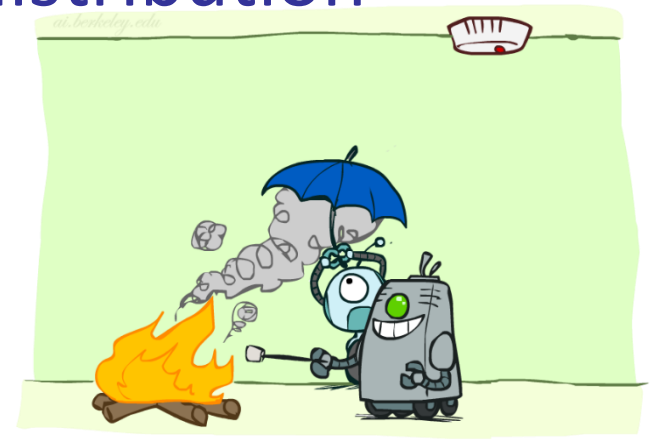
$$\forall x, y \quad P(x, y) = P(x)P(y) \quad \text{---} \rightarrow \quad X \perp Y$$

- X and Y are **conditionally independent** given Z

$$\forall x, y, z \quad P(x, y|z) = P(x|z)P(y|z) \quad \text{---} \rightarrow \quad X \perp Y|Z$$

- (Conditional) independence is a property of a distribution

- Example:  $Alarm \perp Fire|Smoke$

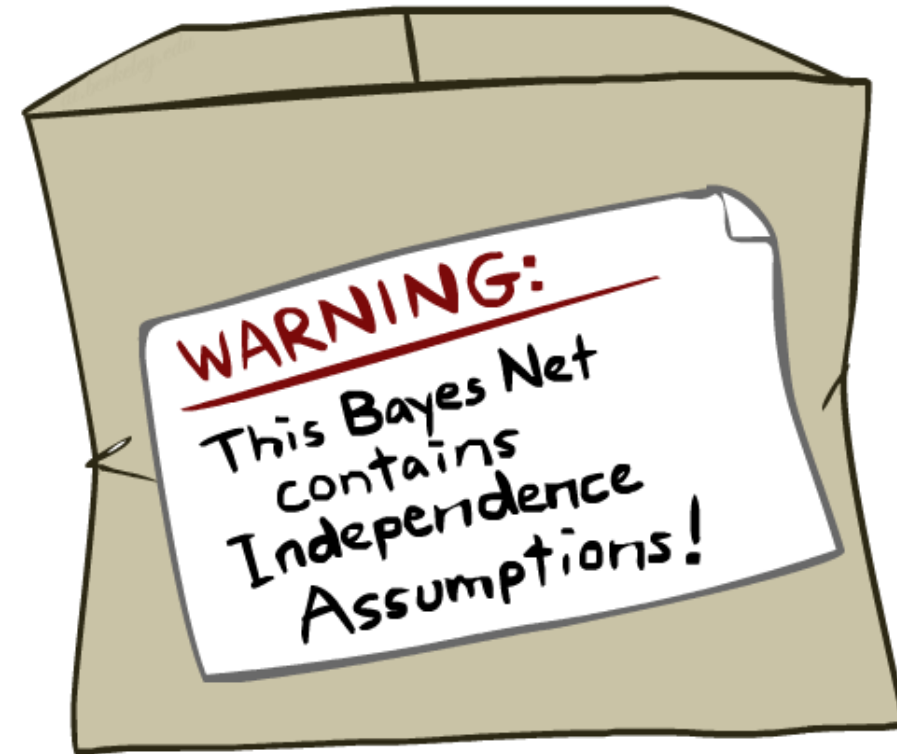


# Bayes Nets: Assumptions

- Assumptions we are required to make to define the Bayes net when given the graph:

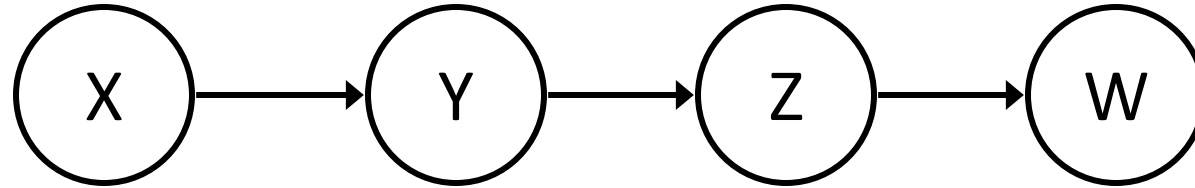
$$P(x_i | x_1 \cdots x_{i-1}) = P(x_i | \text{parents}(X_i))$$

- Beyond above “chain rule → Bayes net” conditional independence assumptions
  - Often additional conditional independences
  - They can be read off the graph
- Important for modeling: understand assumptions made when choosing a Bayes net graph



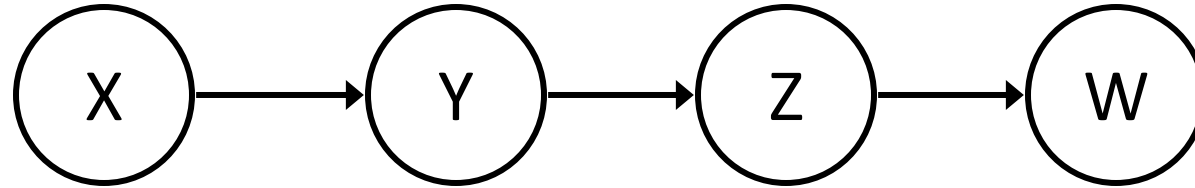
# Example

---



- Conditional independence assumptions directly from simplifications in chain rule:
- Additional implied conditional independence assumptions?

# Example

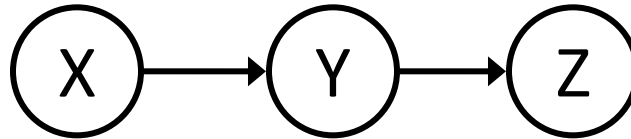


- Conditional independence assumptions directly from simplifications in chain rule:
- Additional implied conditional independence assumptions?

$$\underline{P(W|X,Y)} = \frac{P(W,X,Y)}{P(X,Y)} = \frac{\sum_Z P(X)P(Y|X)P(Z|Y)P(W|Z)}{P(X)P(Y|X)} = \sum_Z P(Z|Y)P(W|Z,Y) = \sum_Z P(W,Z|Y) = \underline{P(W|Y)}$$

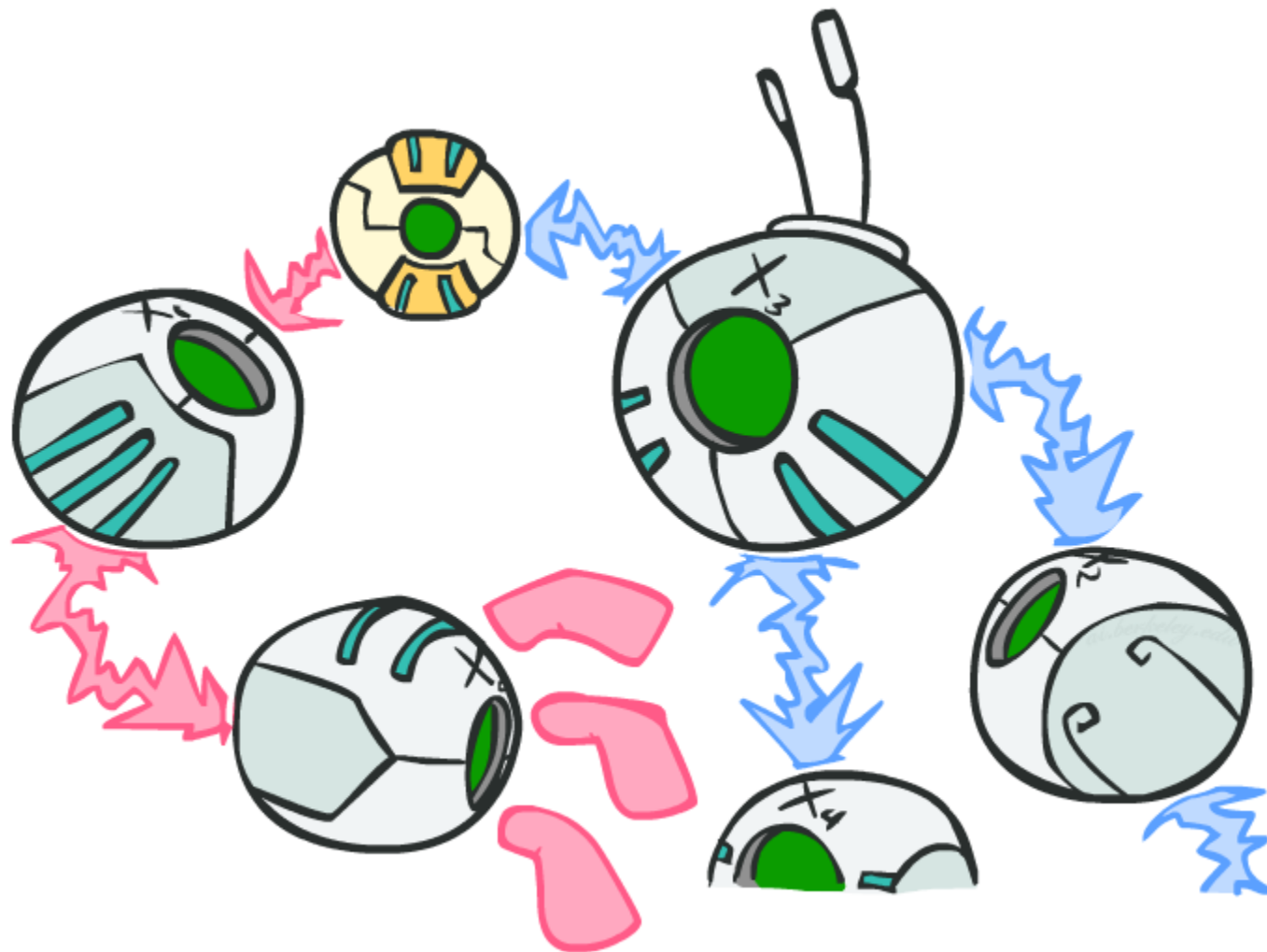
# Independence in a BN

- Important question about a BN:
  - Are two nodes independent given certain evidence?
  - If yes, can prove using algebra (tedious in general)
  - If no, can prove with a counter example
  - Example:



- Question: are X and Z necessarily independent?
  - Answer: no. Example: low pressure causes rain, which causes traffic.
  - X can influence Z, Z can influence X (via Y)
  - Addendum: they *could* be independent: how?

# D-separation: Outline





# D-separation: Outline

---

- Study independence properties for triples
- Analyze complex cases in terms of member triples
- D-separation: a condition / algorithm for answering such queries

# Causal Chains

- This configuration is a “causal chain”



X: Low pressure

Y: Rain

Z: Traffic

$$P(x, y, z) = P(x)P(y|x)P(z|y)$$

- Guaranteed X independent of Z ? **No!**

- One example set of CPTs for which X is not independent of Z is sufficient to show this independence is not guaranteed.

- Example:

- Low pressure causes rain causes traffic, high pressure causes no rain causes no traffic

- In numbers:

$$P(+y \mid +x) = 1, P(-y \mid -x) = 1, \\ P(+z \mid +y) = 1, P(-z \mid -y) = 1$$

# Causal Chains

- This configuration is a “causal chain”

- Guaranteed X independent of Z given Y?



X: Low pressure

Y: Rain

Z: Traffic

$$P(x, y, z) = P(x)P(y|x)P(z|y)$$

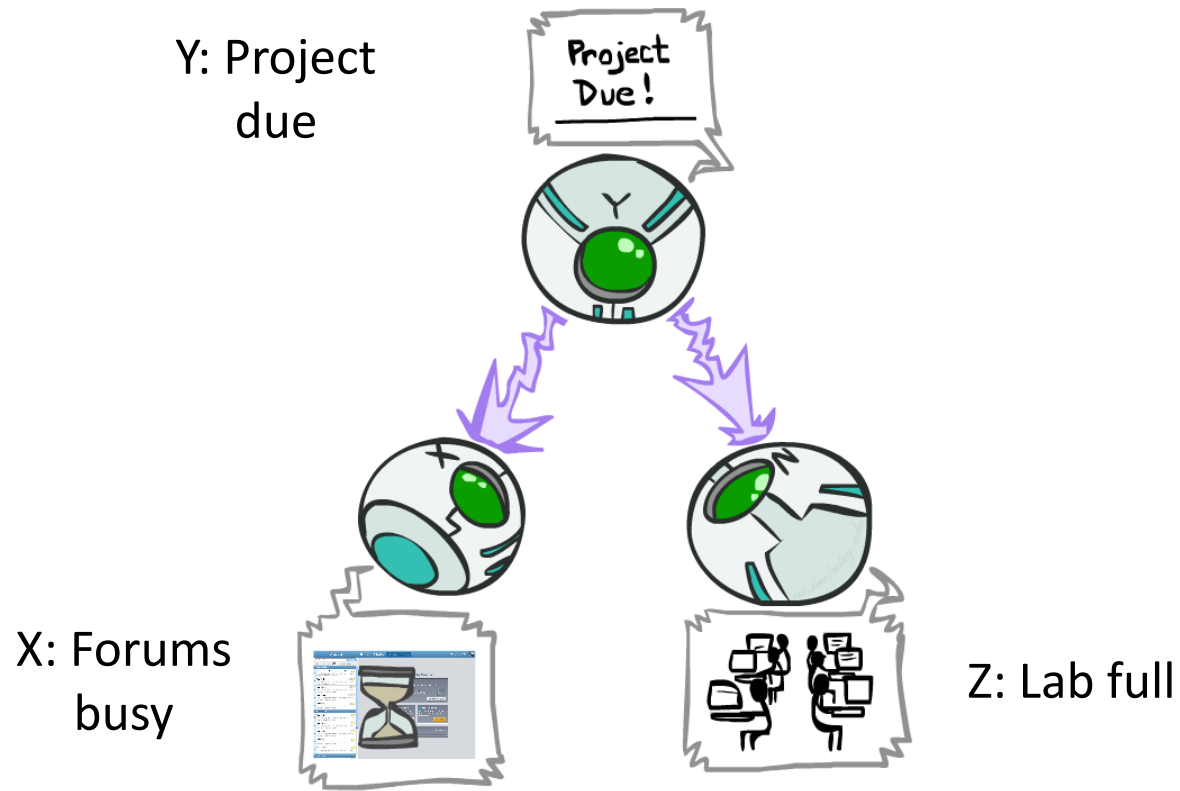
$$\begin{aligned} P(z|x, y) &= \frac{P(x, y, z)}{P(x, y)} \\ &= \frac{P(x)P(y|x)P(z|y)}{P(x)P(y|x)} \\ &= P(z|y) \end{aligned}$$

*Yes!*

- Evidence along the chain “blocks” the influence

# Common Cause

- This configuration is a “common cause”



$$P(x, y, z) = P(y)P(x|y)P(z|y)$$

- Guaranteed X independent of Z ? **No!**

- One example set of CPTs for which X is not independent of Z is sufficient to show this independence is not guaranteed.

- Example:

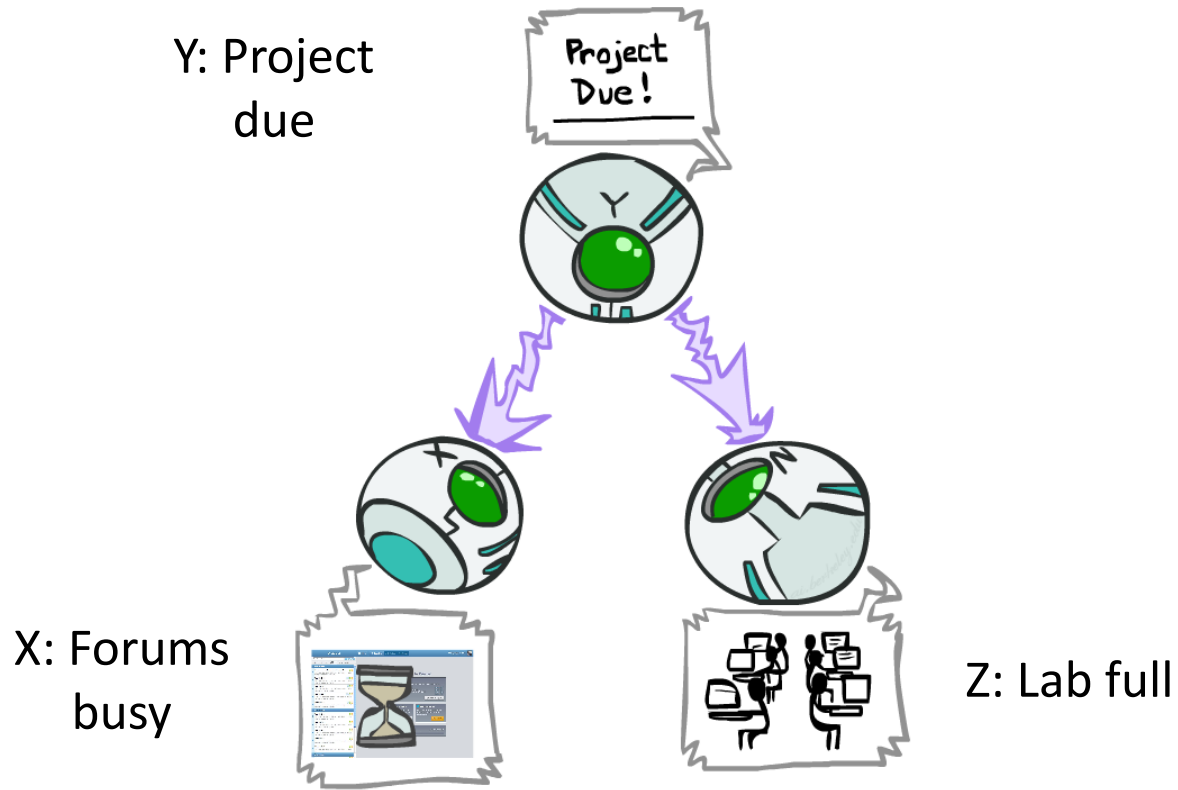
- Project due causes both forums busy and lab full

- In numbers:

$$P(+x \mid +y) = 1, P(-x \mid -y) = 1, \\ P(+z \mid +y) = 1, P(-z \mid -y) = 1$$

# Common Cause

- This configuration is a “common cause”
- Guaranteed X and Z independent given Y?



$$P(x, y, z) = P(y)P(x|y)P(z|y)$$

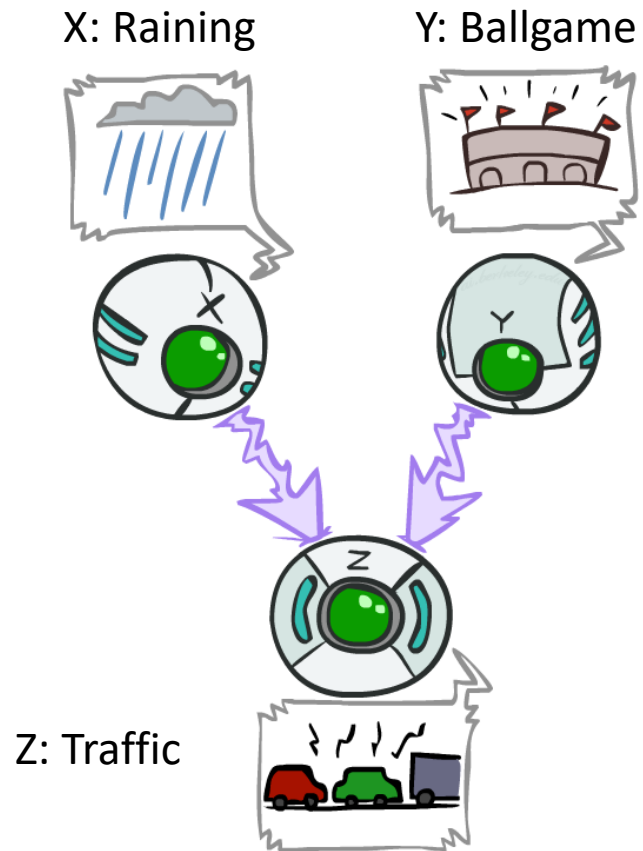
$$\begin{aligned} P(z|x, y) &= \frac{P(x, y, z)}{P(x, y)} \\ &= \frac{P(y)P(x|y)P(z|y)}{P(y)P(x|y)} \\ &= P(z|y) \end{aligned}$$

**Yes!**

- Observing the cause blocks influence between effects.

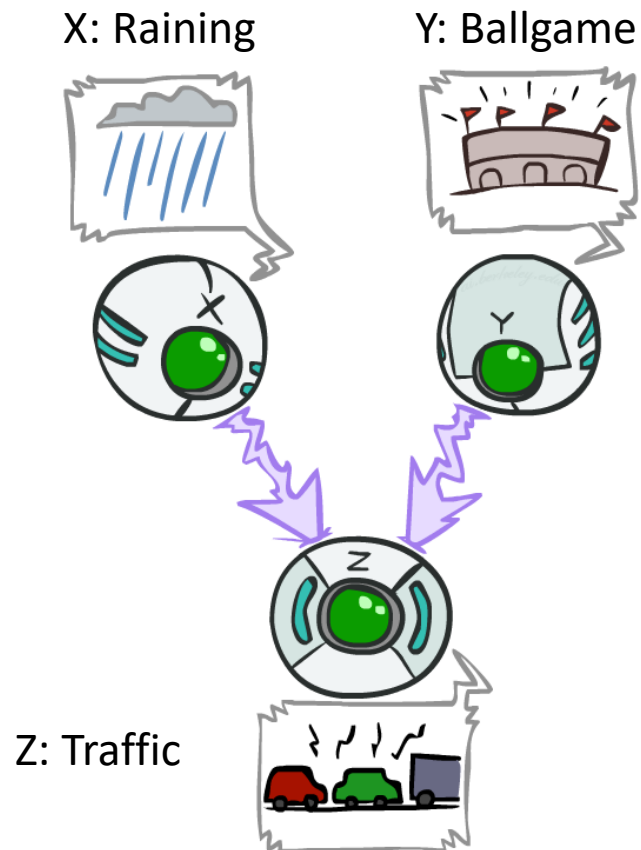
# Common Effect

- Last configuration: two causes of one effect (v-structures)



# Common Effect

- Last configuration: two causes of one effect (v-structures)



- Are X and Y independent?
  - **Yes**: the ballgame and the rain cause traffic, but they are not correlated
  - Still need to prove they must be (try it!)
- Are X and Y independent given Z?
  - **No**: seeing traffic puts the rain and the ballgame in competition as explanation.
- **This is backwards from the other cases**
  - Observing an effect **activates** influence between possible causes.

# The General Case

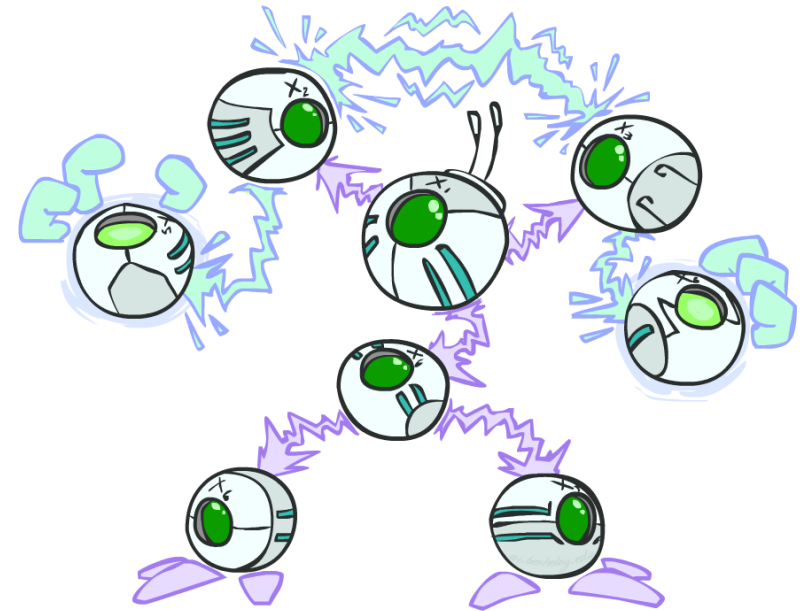
---





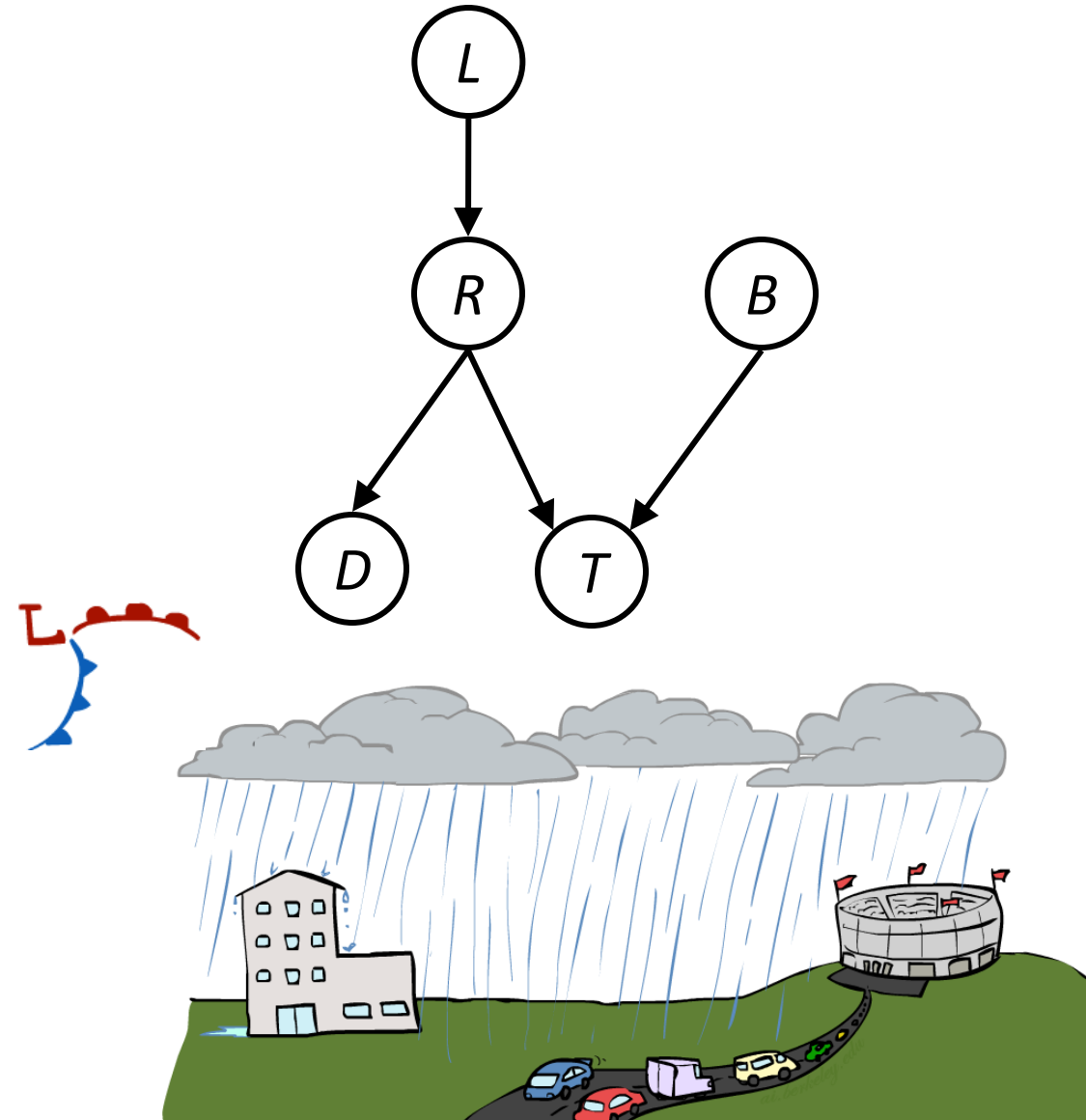
# The General Case

- General question: in a given BN, are two variables independent (given evidence)?
- Solution: analyze the graph
- Any complex example can be broken into repetitions of the three canonical cases



# Reachability

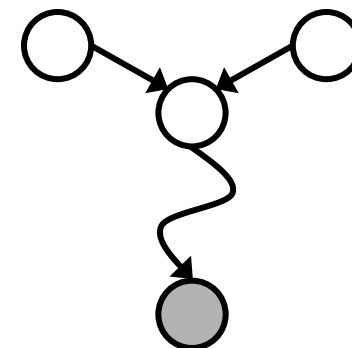
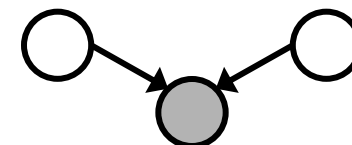
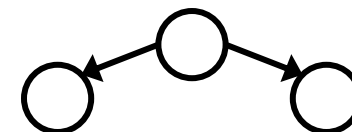
- Recipe: shade evidence nodes, look for paths in the resulting graph
- Attempt 1: if two nodes are connected by an undirected path not blocked by a shaded node, they are conditionally independent
- Almost works, but not quite
  - Where does it break?
  - Answer: the v-structure at T doesn't count as a link in a path unless "active"



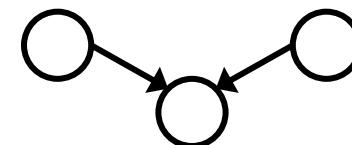
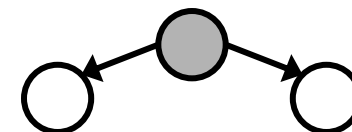
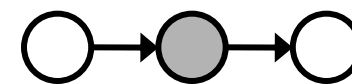
# Active / Inactive Paths

- Question: Are X and Y conditionally independent given evidence variables  $\{Z\}$ ?
  - Yes, if X and Y “d-separated” by Z
  - Consider all (undirected) paths from X to Y
  - No active paths = independence!
- A path is active if each triple is active:
  - Causal chain  $A \rightarrow B \rightarrow C$  where B is unobserved (either direction)
  - Common cause  $A \leftarrow B \rightarrow C$  where B is unobserved
  - Common effect (aka v-structure)  
 $A \rightarrow B \leftarrow C$  where B or one of its descendants is observed
- All it takes to block a path is a single inactive segment

Active Triples



Inactive Triples



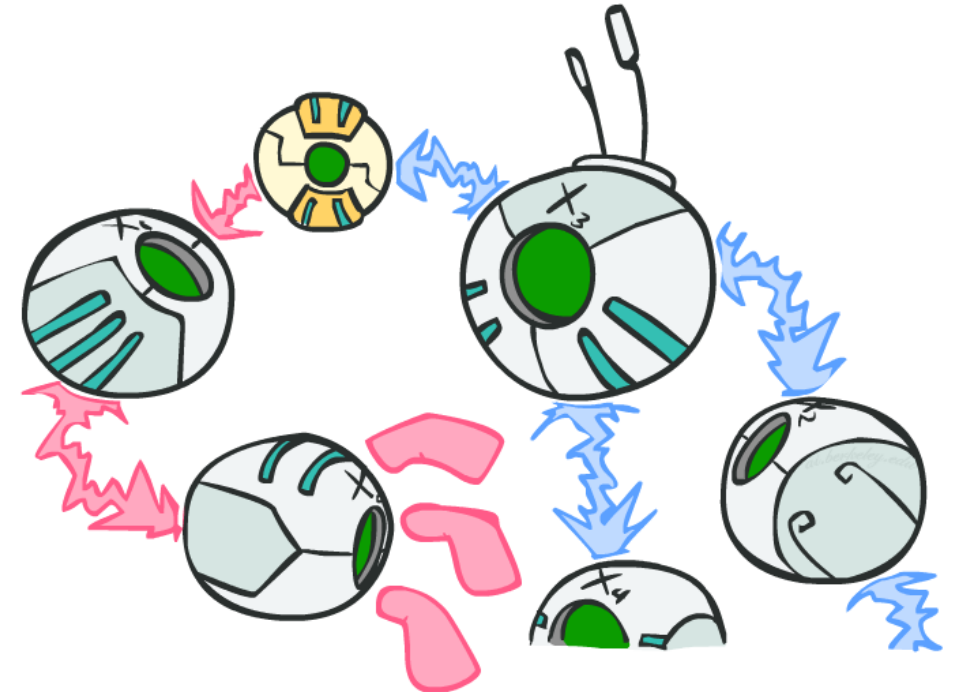
# D-Separation

- Query:  $X_i \perp\!\!\!\perp X_j \mid \{X_{k_1}, \dots, X_{k_n}\} ?$
- Check all (undirected!) paths between  $X_i$  and  $X_j$ 
  - If one or more active, then independence not guaranteed

$$X_i \not\perp\!\!\!\perp X_j \mid \{X_{k_1}, \dots, X_{k_n}\}$$

- Otherwise (i.e. if all paths are inactive), then independence is guaranteed

$$X_i \perp\!\!\!\perp X_j \mid \{X_{k_1}, \dots, X_{k_n}\}$$



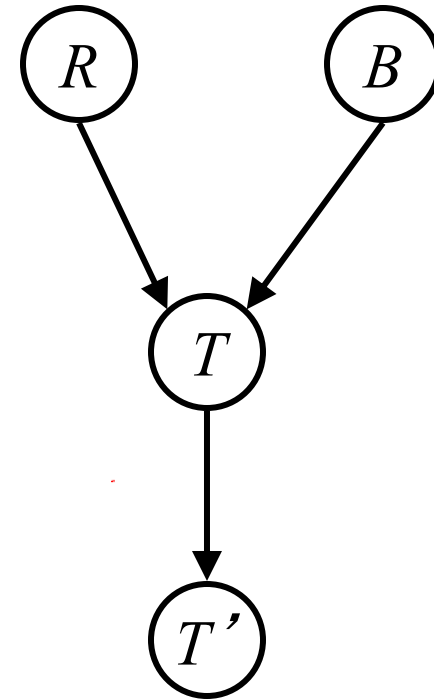
# Example 1

---

$R \perp\!\!\!\perp B$  *Yes*

$R \perp\!\!\!\perp B | T$  *No*

$R \perp\!\!\!\perp B | T'$  *No*



# Example 2

- Variables:

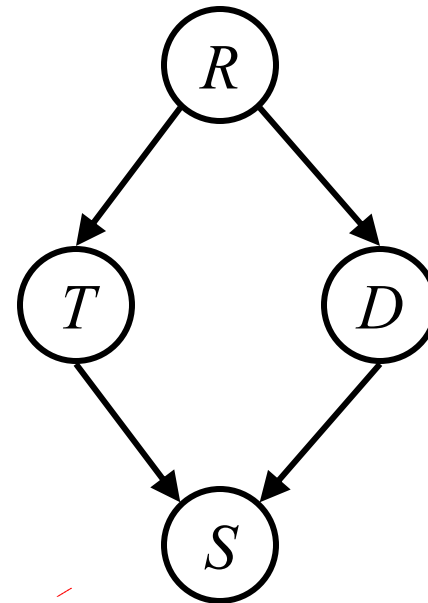
- R: Raining
- T: Traffic
- D: Roof drips
- S: I'm sad

- Questions:

$T \perp\!\!\!\perp D$  *No*

$T \perp\!\!\!\perp D | R$  *Yes*

$T \perp\!\!\!\perp D | R, S$  *No*

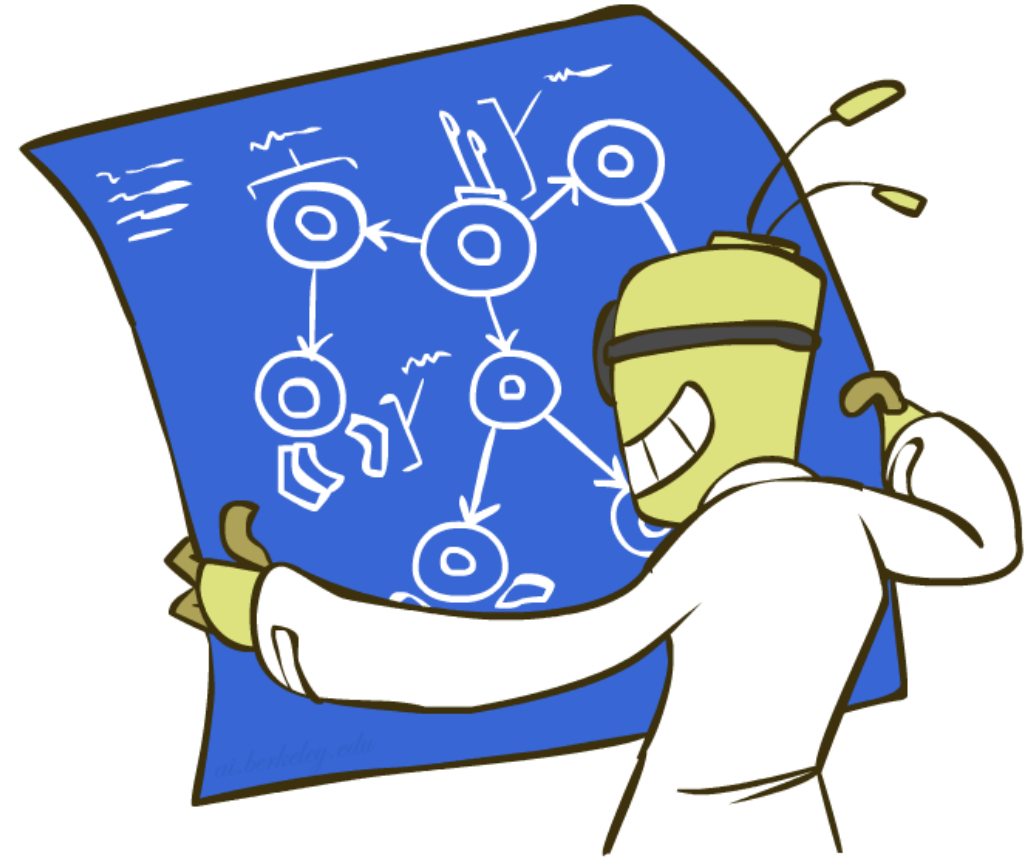


# Structure Implications

- Given a Bayes net structure, can run d-separation algorithm to build a complete list of conditional independences that are necessarily true of the form

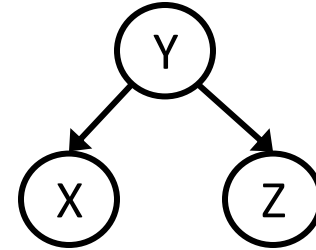
$$X_i \perp\!\!\!\perp X_j | \{X_{k_1}, \dots, X_{k_n}\}$$

- This list determines the set of probability distributions that can be represented

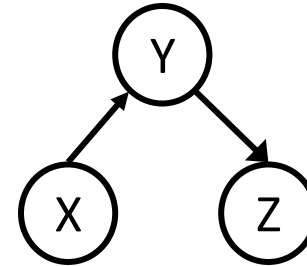


# Computing All Independences

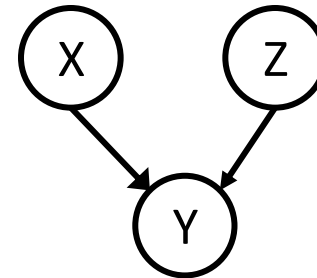
COMPUTE ALL THE  
INDEPENDENCES!



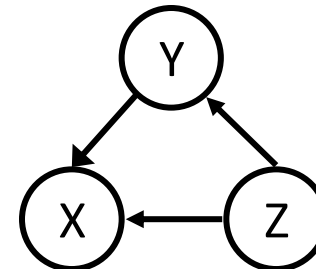
$$X \perp\!\!\!\perp Z \mid Y$$



$$X \perp\!\!\!\perp Z \mid Y$$



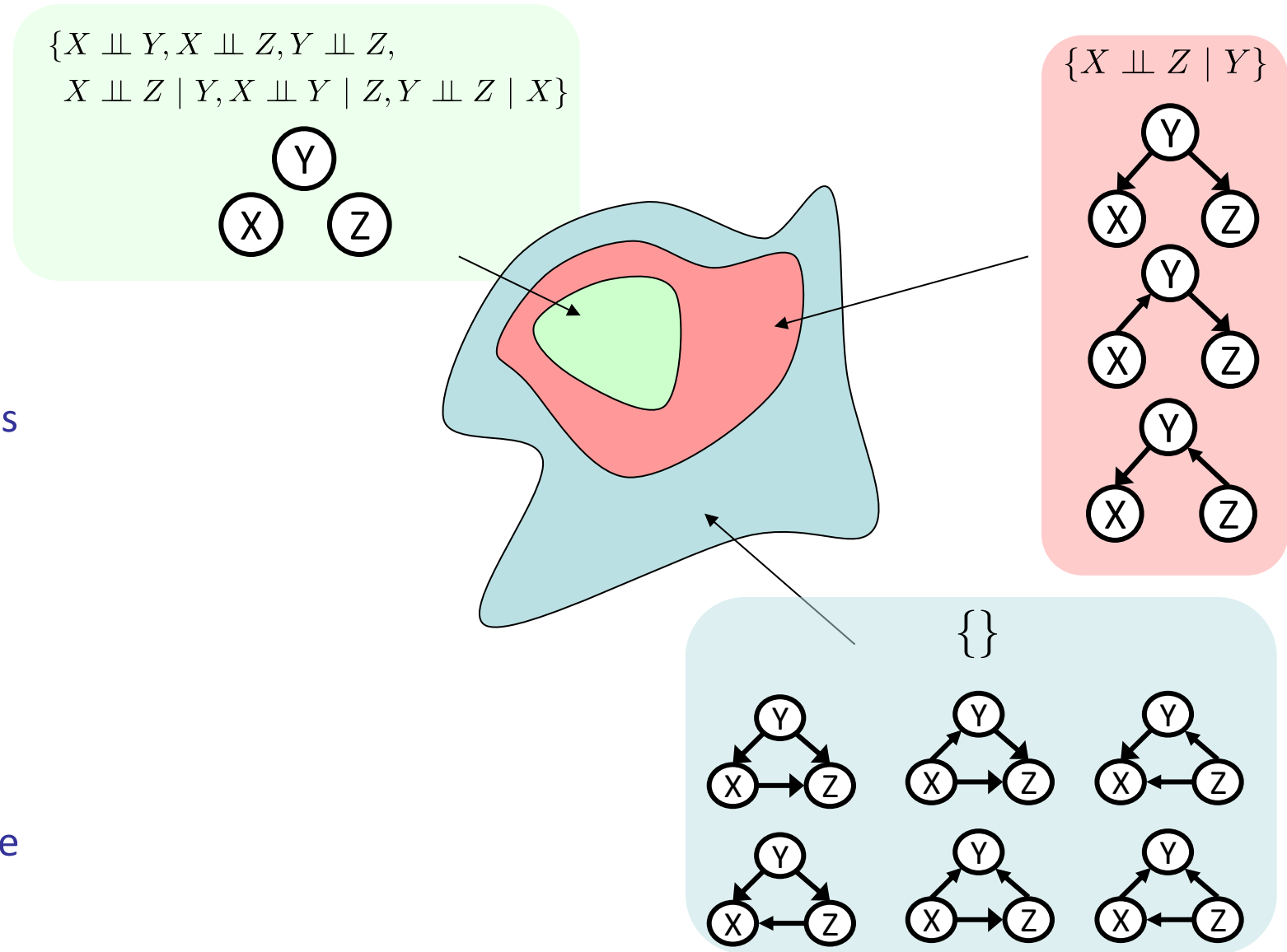
$$X \perp\!\!\!\perp Z$$





# Topology Limits Distributions

- Given some graph topology  $G$ , only certain joint distributions can be encoded
- The graph structure guarantees certain (conditional) independences
- (There might be more independence)
- Adding arcs increases the set of distributions, but has several costs
- Full conditioning can encode any distribution



# Bayes Nets Representation Summary

---

- Bayes nets compactly encode joint distributions
- Guaranteed independencies of distributions can be deduced from BN graph structure
- D-separation gives precise conditional independence guarantees from graph alone
- A Bayes' net's joint distribution may have further (conditional) independence that is not detectable until you inspect its specific distribution

# Bayes' Nets

---



Representation



Conditional Independences

- Probabilistic Inference
  - Enumeration (exact, exponential complexity)
  - Variable elimination (exact, worst-case exponential complexity, often better)
  - Inference is NP-complete
  - Sampling (approximate)
- Learning Bayes' Nets from Data